Patterns among the Primes - *gaps*

https://github.com/fbholt/Primegaps-v2

$\mathcal{G}(p_0^\#) \rightarrow w_g(\lambda)$

Data prep

Cycles of gaps $\mathcal{G}(p^\#)$

Population models $w_g(\lambda)$ for all gaps $g \leq 2p_1$

01_Cycles

02_p-rough

11_lambda

Graphing $w_g(\lambda)$

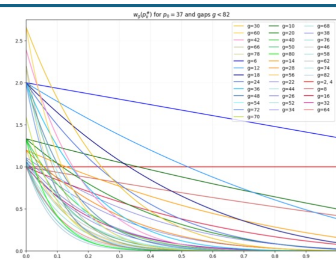
for all gaps $g \leq 2p_1$

$$\lambda(p_0) = 1$$

$$\lambda(p^2) \approx \lambda(p)/2$$

$$\lambda(\infty) = 0$$

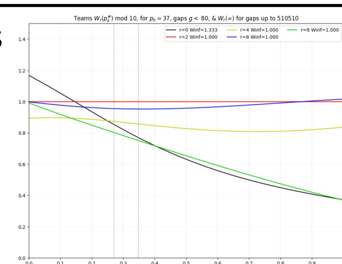
10_wg



Topic: expected biases in last digits

Graphing relative populations $W_r(\lambda)$ by residue *r* for different bases.

16_bias_lastdigits



Intervals of survival $\Delta H(p_k)$

Data prep: Prime gaps in intervals of survival

$$\Delta H(p_k) = (p_k^2, p_{k+1}^2]$$

03_primeblock

12_DelHprep

Graphing $\Delta H(p_k)$

Histogram of counts of prime gaps.

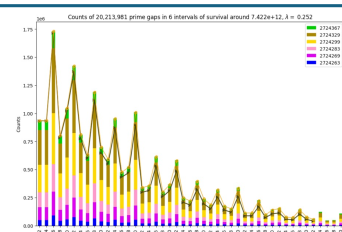
g= 2,4,6,8,10,12,…

Bands color-coded by gap

$g_k = p_{k+1} - p_k$

Compared to estimates based on $w_g(\lambda)$

13_DelHdisplay

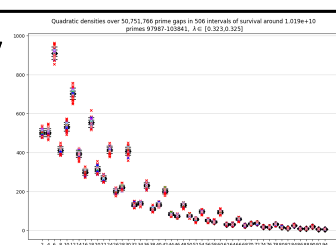


Topic: quadratic density

Graphing the densities of gaps in intervals of survival.

$$\eta_g(p_k) = n_g(\Delta H(p_k))/g_k$$

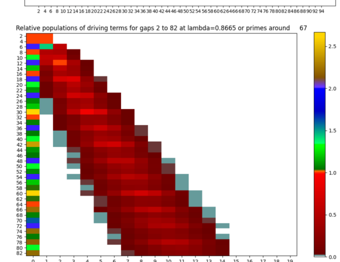
33_quaddensity



Evolution of driving terms

$$w_{g,j}(p_k)$$

34_wgj_evolution



Mar 2026

Patterns among the Primes

admissible constellations

https://github.com/fbholt/Primegaps-v2

$\mathcal{G}(p_0^\#) \rightarrow \; w_s(\lambda)$

Data prep

Population models $w_s(\lambda)$ for select admissible constellations with $|s| \leq 2p_1$

01_Cycles

03_primeblock

Topic:

nonconvex constellations

Studies of Engelsma constellations

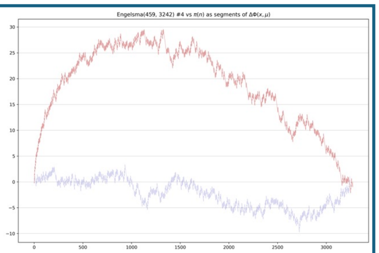
(J, |s|) = (459,3242)

breadth-first searches

23_nonconvex_prep_Engelsma459

depth-first searches

24_nonconvex_search_Engelsma459



Graphing $w_s(\lambda)$

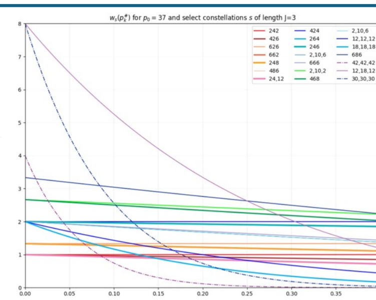
for select constellations of lengths *J=3,5*, with $|s| \leq 2p_1$

$\lambda(p_0) = 1$

$\lambda(p^2) \approx {}^{\lambda(p)}/_{2}$

$\lambda(\infty) = 0$

20_ws



Intervals of survival $\Delta H_s(p_k)$

Data prep

Constellations among primes in intervals of survival $\Delta H(p_k) = (p_k^2, p_{k+1}^2]$

03_primeblock

21_DelHsprep

Graphing $\Delta H(p_k)$

Histogram of counts of constellations among primes.

Bands color-coded by the gap

$g_k = p_{k+1} - p_k$

Compared to estimates based on $w_{s,J}(\lambda)$

22_DelHsdisplay